MAIS PAC

IPhO 02.11: Optics: Lenses, Gratings and Interferometers · Lecture + Problems

Optics: Lenses, Gratings and Interferometers

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

5 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

1. Light as a ray: the limit that works
2. Lenses and image formation
3. Interference and diffraction
4. Interferometers and thin films
5. Problems: reason, calculate, check
6. Further practice and provenance
7. Where to learn more

1. Light as a ray: the limit that works

Geometrical optics is the short-wavelength limit of wave optics. When every object and aperture is much larger than λ, light travels in straight lines in a uniform medium, and the whole subject reduces to two rules: **reflection** (angle in equals angle out) and **refraction**.

At a boundary between media of refractive index n_1 and n_2,

$$n_1\sin\theta_1 = n_2\sin\theta_2,\ v = (c)/(n),\ \lambda_({medium}) = (\lambda_0)/(n). \tag{1}$$

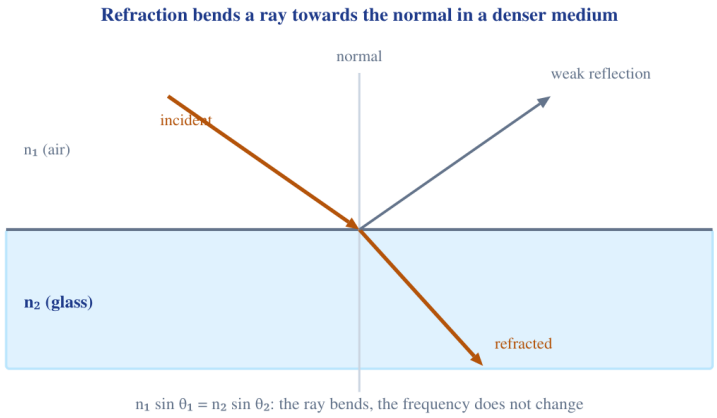


Figure 1. The wavelength shortens inside the denser medium while the frequency stays the same, because it is the frequency that is set by the source. The ray bends because the wave slows down — the marching-band analogy.

The frequency is fixed by the source and does not change when light enters a new medium; the wavelength does. That is the origin of dispersion: n depends slightly on λ, so different colours bend by slightly different amounts and a prism produces a spectrum.

Two consequences are worth memorising:

- **Total internal reflection.** Going from dense to rare, the refracted ray runs along the surface when θ_1 = θ_c with sin θ_c = n_2/n_1. Beyond that angle nothing is transmitted and the surface mirrors perfectly. For glass in air, θ_c≈42°.
- **Apparent depth.** A pool looks shallower than it is by a factor 1/n, because rays from the bottom bend away from the normal as they leave the water.

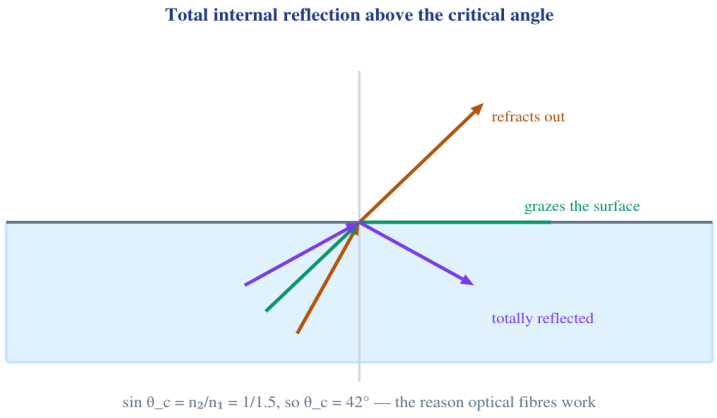


Figure 4. Beyond the critical angle there is no refracted ray at all: the light cannot leave, and the surface becomes a perfect mirror. This is total internal reflection, and it is why a diamond sparkles.

2. Lenses and image formation

A thin lens forms an image according to

$$(1)/(f) = (1)/(u) + (1)/(v), m = (h_i)/(h_o) = -(v)/(u), P = (1)/(f) \{(\text{dioptres, } \} f \{ \text{ in metres})\}. \tag{2}$$

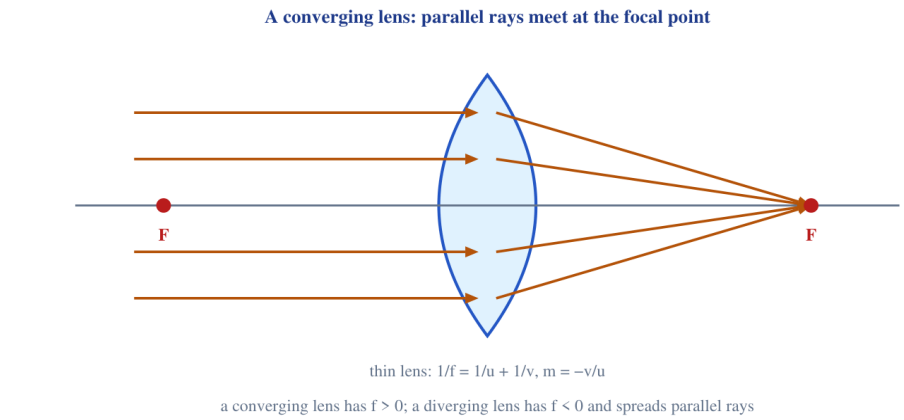


Figure 2. The whole of geometrical optics is the statement that all rays from a point object arrive at one image point. The thin-lens equation is the algebraic version of that picture, with the sign conventions doing the bookkeeping.

The **real-is-positive** convention is the simplest to use: distances measured on the outgoing side of the lens are positive, f>0 for a converging lens, and a negative v means a virtual image on the same side as the object. A negative m means an inverted image.

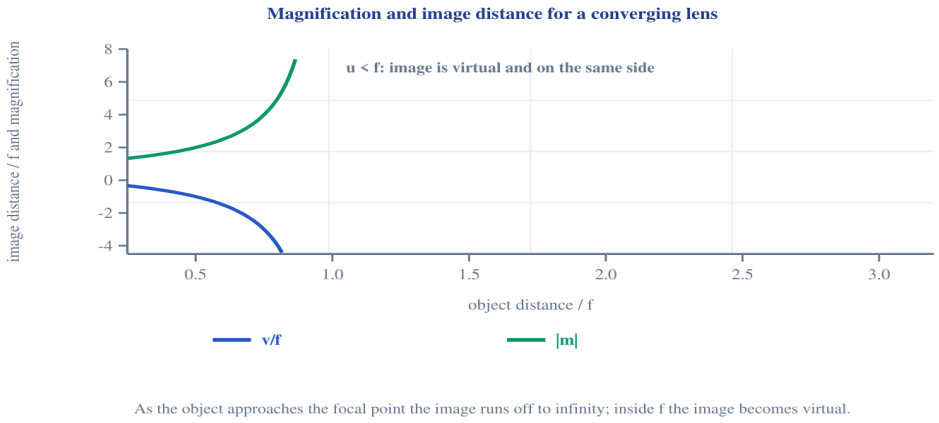


Figure. As the object approaches the focal point the image runs off to infinity; inside f the image becomes virtual.

Learn the six standard cases rather than the algebra:

object position	image	nature
u>2f	between f and 2f	real, inverted, diminished (camera, eye)
u=2f	at 2f	real, inverted, same size
f	beyond 2f	real, inverted, magnified (projector)
u=f	at infinity	parallel rays (collimator)
u	same side, further out	virtual, upright, magnified (magnifier)
diverging lens	same side, closer in	virtual, upright, diminished

Interactive experiment: A thin converging lens — open the online handout to use it.

The **magnifying glass** works in the last row: the image is virtual, so you look **through** the lens, and the angular magnification is M = 25/f (with f in cm) for the relaxed eye.

3. Interference and diffraction

Light is a wave, so two beams can add or cancel. The condition is a path difference:

$$\Delta = d\sin\theta = m\lambda \ \{(\text{bright})\},\ \Delta = (m+(1)/(2))\lambda \ \{(\text{dark})\}. \tag{3}$$

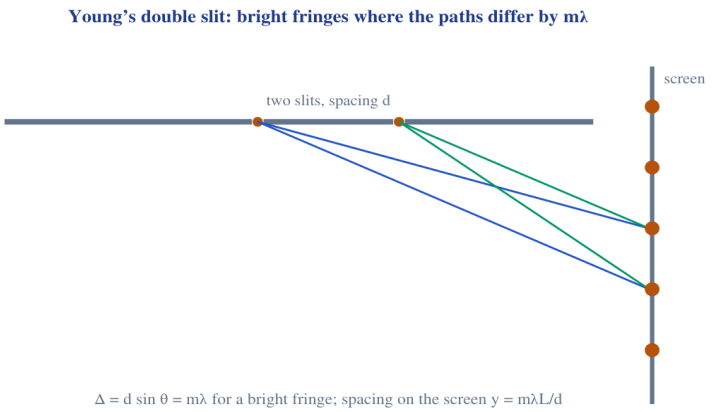


Figure 6. The fringes are equally spaced for small angles, and their spacing gives a way of measuring a wavelength with a ruler — the experiment that settled the wave nature of light.

For a double slit with spacing d observed on a screen a distance L away, the fringe spacing is

$$y = (\lambda\ L)/(d), \tag{4}$$

which is the classic measurement of a wavelength — Young's experiment did exactly this in 1801 and settled the wave theory.

A **diffraction grating** is the same physics with N slits: the maxima are at the same angles, dsin θ = mλ, but they become N times narrower and N² times brighter as N grows. That is why gratings, not prisms, are the standard tool in spectroscopy.

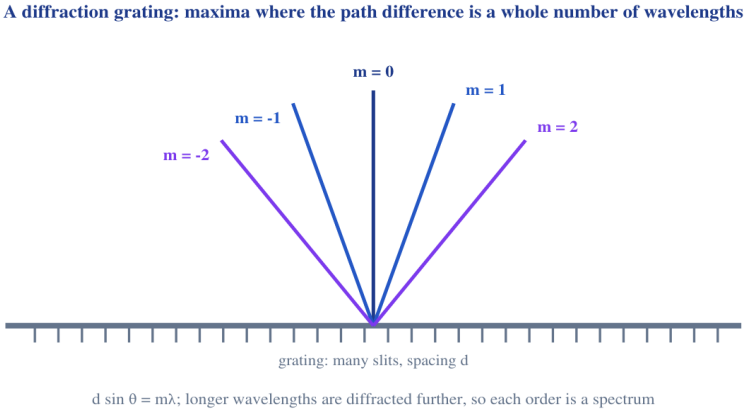


Figure 5. With N slits the maxima become N times narrower and N² times brighter, which is why gratings, not prisms, are used to measure wavelengths precisely.

Single-slit diffraction gives a broad central maximum of half-width θ≈λ/a for a slit of width a, with weaker secondary maxima on each side. This sets the **resolution limit** of any optical instrument:

$$\theta_(\text{min}) \approx 1.22(\lambda)/(D) \tag{5}$$

for a circular aperture of diameter D — the Rayleigh criterion, and the reason a telescope's aperture, not its magnification, decides how much detail it can show.

4. Interferometers and thin films

An interferometer splits a beam, sends the two halves along different paths, and recombines them. A change of path length of λ/2 flips a bright fringe to a dark one, so counting fringes measures distances to a fraction of a wavelength — the basis of the Michelson interferometer, of gravitational-wave detectors, and of optical coherence.

$$2nt = (m+(1)/(2))λ \ \{(\text{bright reflection from a film of thickness } \} \ t\{)\} \tag{6}$$

Thin films show colours because the two reflected waves have a path difference set by 2nt; a soap bubble's colours change with thickness, and an anti-reflection coating is designed with nt = λ/4 so the two reflections cancel.

The optics checklist. Draw the ray diagram first, always — it catches sign errors before they happen → for refraction, decide which side is denser and whether total internal reflection is possible → for lenses, use 1/f=1/u+1/v with the real-is-positive convention, then check the answer against the table of six cases → for interference, the whole problem is the path difference → for gratings, dsin θ=mλ with d the **slit spacing**, not the slit width → check the small-angle approximation sin θ≈tan θ≈θ whenever y≪ L.

5. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Refraction into glass. A ray of light strikes a glass block of refractive index 1.50 at 40° to the normal. (a) Find the angle of refraction. (b) Find the speed and wavelength of the light inside, given λ_0 = 600 {nm}. (c) Find the critical angle for light trying to leave the glass.

Worked solution →

[1] Problem 2 — A converging lens. An object 3.0 {cm} tall stands 40 {cm} from a converging lens of focal length 15 {cm}. (a) Find the image distance. (b) Find the magnification and the image height. (c) Describe the image and state one device that uses this arrangement.

Worked solution →

[2] Problem 3 — The magnifying glass. A converging lens of focal length 8.0 {cm} is used as a magnifier, with the object 6.0 {cm} from the lens. (a) Find the image distance and magnification. (b) State whether the image is real or virtual and explain how you would view it. (c) Find the angular magnification for a relaxed eye focused at infinity.

Worked solution →

[2] Problem 4 — Total internal reflection and optical fibres. An optical fibre has a core of index 1.52 surrounded by cladding of index 1.48. (a) Find the critical angle at the core–cladding boundary. (b) Find the maximum angle to the fibre axis at which light can enter the end face (air, n=1.00) and still be guided. (c) Explain why the fibre must be thin as well as having a suitable index difference.

Worked solution →

[2] Problem 5 — Young's double slit. Light of wavelength 589 {nm} passes through two slits 0.40 {mm} apart, and the fringes are observed on a screen 2.0 {m} away. (a) Find the fringe spacing. (b) Find the distance from the centre to the third bright fringe. (c) Find the path difference at that point in wavelengths.

Worked solution →

[2] Problem 6 — A diffraction grating. A grating with 500 lines per millimetre is illuminated by light of wavelength 600 {nm} at normal incidence. (a) Find the slit spacing. (b) Find the angle of the first and second order maxima. (c) Show that no third order exists, and explain what happens at the largest possible order.

Worked solution →

[2] Problem 7 — Two lenses in contact. A converging lens of focal length 20 {cm} and a diverging lens of focal length -30 {cm} are placed in contact. (a) Find the focal length and power of the combination. (b) An object is placed 50 {cm} from the combination: find the image position and magnification. (c) Explain why opticians add powers rather than focal lengths.

Worked solution →

[3] Problem 8 — A thin film. A soap film of refractive index 1.33 is illuminated at near-normal incidence by white light. It appears green (λ = 550 {nm}) in reflected light. (a) Find the minimum thickness for constructive interference. (b) Find the two wavelengths in the visible range that would also be reflected strongly at that thickness. (c) Explain why the film looks black just before it bursts.

Worked solution →

[3] Problem 9 — The Rayleigh criterion. A telescope has an aperture of diameter 0.20 {m}. (a) Find the angular resolution at λ = 550 {nm}. (b) Find the smallest crater it can resolve on the Moon, 3.8×10^8 {m} away. (c) Explain why increasing the magnification does not improve the resolution, and what a larger aperture does.

Worked solution →

[3] Problem 10 — A prism at minimum deviation. A prism of apex angle 60° and refractive index 1.50 is used at minimum deviation. (a) Show that at minimum deviation the ray passes symmetrically, with r_1 = r_2 = A/2. (b) Find the angle of minimum deviation. (c) Explain why the prism spreads white light into a spectrum and why a grating does it better.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

MIT OpenCourseWare 8.03, Vibrations and Waves

Oscillators, resonance, superposition and optics with demonstrations.